HOMEWORK SET 3

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1. Dynamic Array – Largest Element

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n, i, max;
    int *a;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    a = malloc(n * sizeof(int));

    printf("Enter elements: ");
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    max = a[0];
    for (i = 1; i < n; i++)
        if (a[i] > max)
            max = a[i];

    printf("Largest element = %d\n", max);

    free(a);
    return 0;
}
```

Sample Output

```
Terminal Output

Enter number of elements: 6
Enter elements: 18 42 7 91 35 64
Largest element = 91
```

2. Reverse an Array Using Pointers

```c
#include <stdio.h>

int main() {
    int a[100], n, i;
    int *ptr;

    printf("Enter size: ");
    scanf("%d", &n);

    printf("Enter elements: ");
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    ptr = &a[n - 1];

    printf("Reverse order: ");
    for (i = 0; i < n; i++)
        printf("%d ", *ptr--);

    return 0;
}
```

Sample Output

Terminal Output

```
Enter size: 5
Enter elements: 12 25 38 41 56
Reverse order: 56 41 38 25 12
```

3. Student Records Sorted by Name

```
#include <stdio.h>
#include <string.h>

struct Student {
    int roll;
    char name[40];
    float marks;
};

int main() {
    struct Student s[50], temp;
    int n, i, j;

    printf("Number of students: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        printf("Enter roll, name and marks: ");
        scanf("%d %s %f", &s[i].roll, s[i].name, &s[i].marks);
    }

    for (i = 0; i < n - 1; i++)
        for (j = i + 1; j < n; j++)
            if (strcmp(s[i].name, s[j].name) > 0) {
                temp = s[i];
                s[i] = s[j];
                s[j] = temp;
            }

    printf("\nStudents in name order:\n");
    for (i = 0; i < n; i++)
        printf("%d  %s  %.2f\n", s[i].roll, s[i].name, s[i].marks);

    return 0;
}
```

Sample Output

```
Terminal Output
```

```
Number of students: 3
Enter roll, name and marks: 12 Rameez 88.5
Enter roll, name and marks: 7 Arjun 91.0
Enter roll, name and marks: 19 Zoya 84.5

Students in name order:
7  Arjun  91.00
12  Rameez  88.50
19  Zoya  84.50
```

4. Add Distances in Feet-Inch Format

```c
#include <stdio.h>

struct Distance {
    int feet;
    int inch;
};

int main() {
    struct Distance d[20], total = {0, 0};
    int n, i;

    printf("Number of distances: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        printf("Distance %d (feet inches): ", i + 1);
        scanf("%d %d", &d[i].feet, &d[i].inch);
        total.feet += d[i].feet;
        total.inch += d[i].inch;
    }

    total.feet += total.inch / 12;
    total.inch %= 12;

    printf("Total = %d feet %d inches\n", total.feet, total.inch);
    return 0;
}
```

Sample Output

Terminal Output

```
Number of distances: 3
Distance 1 (feet inches): 5 8
Distance 2 (feet inches): 7 11
Distance 3 (feet inches): 3 6
Total = 17 feet 1 inches
```

5. Add Two Complex Numbers Using a Function

```c
#include <stdio.h>

struct Complex {
    float real, imag;
};

struct Complex sum(struct Complex x, struct Complex y) {
    struct Complex z;
    z.real = x.real + y.real;
    z.imag = x.imag + y.imag;
    return z;
}

int main() {
    struct Complex a, b, result;

    printf("First complex number (real imag): ");
    scanf("%f %f", &a.real, &a.imag);

    printf("Second complex number (real imag): ");
    scanf("%f %f", &b.real, &b.imag);

    result = sum(a, b);

    printf("Sum = %.2f + %.2fi\n", result.real, result.imag);
    return 0;
}
```

Sample Output

Terminal Output

```
First complex number (real imag): 4.5 2.5
Second complex number (real imag): 3.0 6.5
Sum = 7.50 + 9.00i
```

6. Inventory Management Using Structure

```
#include <stdio.h>

struct Item {
    int code;
    char name[30];
    int qty;
    float price;
};

int main() {
    struct Item item[50];
    int n, i;

    printf("Number of items: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        printf("Enter code, name, quantity and price: ");
        scanf("%d %s %d %f", &item[i].code, item[i].name,
              &item[i].qty, &item[i].price);
    }

    printf("\nInventory:\n");
    for (i = 0; i < n; i++)
        printf("%d  %-12s Qty:%d  Price:%.2f\n",
               item[i].code, item[i].name,
               item[i].qty, item[i].price);

    return 0;
}
```

Sample Output

Terminal Output

```
Number of items: 2
Enter code, name, quantity and price: 101 Keyboard 8 799.50
Enter code, name, quantity and price: 205 Mouse 15 449.00

Inventory:
101  Keyboard     Qty:8  Price:799.50
205  Mouse        Qty:15  Price:449.00
```

7. Decimal to Binary, Octal and Hexadecimal

```c
#include <stdio.h>

void binary(int n) {
    if (n > 1)
        binary(n / 2);
    printf("%d", n % 2);
}

int main() {
    int n;

    printf("Enter decimal number: ");
    scanf("%d", &n);

    printf("Binary = ");
    if (n == 0) printf("0");
    else binary(n);

    printf("\nOctal = %o", n);
    printf("\nHexadecimal = %X\n", n);

    return 0;
}
```

Sample Output

Terminal Output

```
Enter decimal number: 156
Binary = 10011100
Octal = 234
Hexadecimal = 9C
```

8. Difference Between Two Time Periods

```c
#include <stdio.h>

struct Time {
    int h, m, s;
};

int main() {
    struct Time t1, t2, ans;
    int x, y, diff;

    printf("Enter first time (HH MM SS): ");
    scanf("%d %d %d", &t1.h, &t1.m, &t1.s);

    printf("Enter second time (HH MM SS): ");
    scanf("%d %d %d", &t2.h, &t2.m, &t2.s);

    x = t1.h * 3600 + t1.m * 60 + t1.s;
    y = t2.h * 3600 + t2.m * 60 + t2.s;

    diff = x > y ? x - y : y - x;

    ans.h = diff / 3600;
    diff %= 3600;
    ans.m = diff / 60;
    ans.s = diff % 60;

    printf("Difference = %02d:%02d:%02d\n", ans.h, ans.m, ans.s);
    return 0;
}
```

Sample Output

Terminal Output

```
Enter first time (HH MM SS): 14 35 20
Enter second time (HH MM SS): 09 48 55
Difference = 04:46:25
```

9. Employee Management Using Structure

```c
#include <stdio.h>

struct Employee {
    int id;
    char name[40];
    float salary;
};

int main() {
    struct Employee e[50];
    int n, i;

    printf("Number of employees: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        printf("Enter ID, name and salary: ");
        scanf("%d %s %f", &e[i].id, e[i].name, &e[i].salary);
    }

    printf("\nEmployee Details:\n");
    for (i = 0; i < n; i++)
        printf("ID:%d  Name:%s  Salary:%.2f\n",
               e[i].id, e[i].name, e[i].salary);

    return 0;
}
```

Sample Output

Terminal Output

```
Number of employees: 3
Enter ID, name and salary: 501 Ayaan 42000
Enter ID, name and salary: 502 Meera 46500
Enter ID, name and salary: 503 Rohan 39800

Employee Details:
ID:501  Name:Ayaan  Salary:42000.00
ID:502  Name:Meera  Salary:46500.00
ID:503  Name:Rohan  Salary:39800.00
```

10. Area and Perimeter Using Function Pointers

```c
#include <stdio.h>

float rect_area(float x, float y) { return x * y; }
float rect_peri(float x, float y) { return 2 * (x + y); }
float tri_area(float x, float y) { return 0.5f * x * y; }

int main() {
    float (*operation)(float, float);
    float a, b;

    printf("Rectangle length and breadth: ");
    scanf("%f %f", &a, &b);

    operation = rect_area;
    printf("Rectangle area = %.2f\n", operation(a, b));

    operation = rect_peri;
    printf("Rectangle perimeter = %.2f\n", operation(a, b));

    printf("\nTriangle base and height: ");
    scanf("%f %f", &a, &b);

    operation = tri_area;
    printf("Triangle area = %.2f\n", operation(a, b));

    return 0;
}
```

Sample Output

Terminal Output

```
Rectangle length and breadth: 12 7
Rectangle area = 84.00
Rectangle perimeter = 38.00

Triangle base and height: 10 6
Triangle area = 30.00
```